

typcas v2 Unified Surface Test

1. Strict Regression Gates

All strict gates passed.

2. UX-First Usage

context simplify (expr-of):

Before:

$$\sin(x)^2 + \cos(x)^2$$

After:

$$1$$

context diff (expr-of):

Before:

$$x^3 + \ln(x)$$

After:

$$\frac{1}{x} + 3x^2$$

context solve roots (roots-of):

Before:

$$x^2 - 4$$

After:

After (roots):

$$r_1 = 2$$

$$r_2 = -2$$

After (count):

$$2$$

context error message helper (on success):

none

context trace-integrate:

$\int 2x \cdot \cos(x^2) \, dx$
set $u_1 = x^2$
derive $du_1 = x \cdot 2$
 $= \int 2x \cdot \cos(x^2) \, dx \Rightarrow \int \cos(u_1) \, du_1$ (Apply u-substitution)
 $= \int \cos(u_1) \, du_1 \mapsto \sin(u_1) + C$ (Integrate in u_1)
 $= \sin(u_1) + C \Rightarrow \sin(x^2) + C$ (Back-substitute)
Active assumptions panel: active=0, satisfied=0, conflicts=0.

3. Canonical API Surface (cas.*)

parse/display:

$$\log_2(x) + \sin(x)^2 + \cos(x)^2$$

equation helper:

Before:

$$x^2 - 1$$

After:

$$(x-1)\cdot (x+1)$$

simplify: Before:

$$\log_2(x)+\sin(x)^2+\cos(x)^2$$

After:

$$\log_2(x)+1$$

restriction panel (simplify):

active=0, satisfied=3, conflicts=0 satisfied “:”

$$2>0$$

satisfied “:”

$$2\neq 1$$

satisfied “:”

$$x>0$$

parse-domain input: $(-\infty,0)\cup(0,\infty)$ parse-domain (normalized display):

$$(-\infty,0)\cup(0,\infty)$$

display-domain (a-pos):

$$(0,\infty)$$

query domain value:

$$(0,\infty)$$

diff(order=2)

Before:

$$x^3+\ln(x)$$

After:

$$\frac{-1}{x^2}+6x$$

integrate

Before:

$$x^2+1$$

After:

$$\frac{1}{3}\cdot x^3+x+C$$

definite integrate [0,2]

Before:

$$x$$

After:

$$2$$

implicit diff

Before:

$$x\cdot y-3$$

After:

$$-\left(\frac{y}{x}\right)$$

solve roots count (distinct):

Before:

$$x^2-4$$

After:

$$2$$

factor

Before:

$$x^2-4$$

After:

$$(x-2)\cdot (x+2)$$

taylor

Before:

$$e^x$$

After:

$$\frac{1}{24}\cdot x^4+\frac{1}{6}\cdot x^3+\frac{1}{2}\cdot x^2+x+1$$

limit

Before:

$$\frac{\sin(x)}{x}$$

After:

$$1$$

limit (left-sided)

Before:

$$\frac{\sin(x)}{x},x\rightarrow 0^-$$

After:

$$1$$

limit (right-sided)

Before:

$$\frac{\sin(x)}{x},x\rightarrow 0^+$$

After:

$$1$$

limit (trig standard)

Before:

$$\frac{1-\cos(x)}{x^2}$$

After:

$$\frac{1}{2}$$

limit (x -> inf, rational degree)
Before:

$$\frac{3x^2 + 1}{2x^2 - 5}, x \rightarrow \infty$$

After:

$$1.5$$

eval:
Before:

$$x^2 + 1$$

After:

$$10$$

substitute
Before:

$$x^2 + y, x \rightarrow z + 1$$

After:

$$(z + 1)^2 + y$$

$\int 2x \cdot \cos(x^2) \, \mathrm{d} x$
 set $u_1 = x^2$
 derive $du_1 = x \cdot 2$
 $= \int 2x \cdot \cos(x^2) \, \mathrm{d} x \Rightarrow \int \cos(u_1) \, \mathrm{d} u_1$ (Apply u-substitution)
 $= \int \cos(u_1) \, \mathrm{d} u_1 \mapsto \sin(u_1) + C$ (Integrate in u_1)
 $= \sin(u_1) + C \Rightarrow \sin(x^2) + C$ (Back-substitute)
Active assumptions panel: active=0, satisfied=0, conflicts=0.

4. Query Input + Task API (No Field-Call Parens)

query parsed:

$$x + x + 1$$

query simplify
Before:

$$x + x + 1$$

After:

$$2x + 1$$

query diff
Before:

$$x + x + 1$$

After:

$$2$$

query integrate
Before:

$$x + x + 1$$

After:

$$x^2 + x + C$$

query implicit diff

Before:

$$x^2 + y^2 - 1$$

After:

$$-\left(\frac{x \cdot 2}{y \cdot 2}\right)$$

query solve roots (distinct):

Before:

$$x + x + 1$$

After:

$$1$$

query recreated context (with equivalent):

real / strict= **true**

query domain:

$$(-\text{inf}, \text{inf})$$

task chain diff -> simplify

Before:

$$2$$

After:

$$2$$

$$\begin{aligned} & \frac{d}{dx}(x + x + 1) \\ &= \frac{d}{dx}(2x + 1) \Rightarrow \frac{d}{dx}(2x) + \frac{d}{dx}(1) \quad (\text{Apply sum rule}) \\ & \hookrightarrow \text{Branch 1} \\ &= \frac{d}{dx}(x \cdot 2) \Rightarrow \frac{d}{dx}(x) \cdot 2 + x \cdot \frac{d}{dx}(2) \quad (\text{Apply product rule}) \\ & \hookrightarrow \text{Branch 1} \\ &= \frac{d}{dx}(x) \Rightarrow 1 \quad (\text{Differentiate}) \\ & \hookrightarrow \text{Branch 2} \\ &= \frac{d}{dx}(2) \Rightarrow 0 \quad (\text{Differentiate}) \\ &= \frac{d}{dx}(x) \cdot 2 + x \cdot \frac{d}{dx}(2) \mapsto 2 \quad (\text{Simplify derivative}) \\ & \hookrightarrow \text{Branch 2} \\ &= \frac{d}{dx}(1) \Rightarrow 0 \quad (\text{Differentiate}) \\ &= \frac{d}{dx}(2x) + \frac{d}{dx}(1) \mapsto 2 \quad (\text{Simplify derivative}) \\ & \text{Active assumptions panel: active}=0, \text{ satisfied}=0, \text{ conflicts}=0. \end{aligned}$$

5. Matrix / Systems / Poly Task APIs

matrix add

Before:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

After:

$$\begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

matrix sub
Before:

$$\begin{pmatrix} 5 & 7 \\ 9 & 11 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

After:

$$\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix}$$

matrix scale
Before:

$$2 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

After:

$$\begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$$

matrix mul
Before:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

After:

$$\begin{pmatrix} 4 & 4 \\ 10 & 8 \end{pmatrix}$$

matrix transpose
Before:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

After:

$$\begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

matrix det
Before:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

After:

$$-2$$

matrix inv
Before:

$$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$$

After:

$$\begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

matrix solve

Before (A):

$$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

Before (b):

$$b = (5, 6)$$

After:

$$x_1 = \frac{9}{5}$$

$$x_2 = \frac{7}{5}$$

matrix eigenvalues:

Before:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

After:

$$\lambda_1 = 3$$

$$\lambda_2 = 1$$

matrix eigenvectors:

Before:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

After:

$$\lambda_1 = 3, v_1 = (1, 1)$$

$$\lambda_2 = 1, v_2 = (1, -1)$$

linear system:

Before:

$$2x + y - 5, x - y - 1$$

After:

$$x = 2, y = 1$$

nonlinear system:

Before:

$$x^2 + y^2 - 5, x - y - 1$$

After:

$$\text{converged} = \text{true}, \text{iterations} = 1$$

$$x \approx 2, y \approx 1$$

poly div:

Before:

$$\frac{x^3-1}{x-1}$$

After:

$$q(x)=x^2+x+1,r(x)=0$$

partial fractions:

Before:

$$\frac{2x+3}{x^2-1}$$

After:

$$\frac{\frac{-1}{2}}{x+1}+\frac{\frac{5}{2}}{x-1}$$

6. Unified Demo Flow

Domain (x):

$$(0,\mathrm{inf})$$

demo simplify

Before:

$$\frac{x+1}{x+1}+\ln(e^x)+\sin(x)^2+\cos(x)^2+\sqrt{x^2}+\frac{2}{5}+\frac{7}{15}-\frac{13}{15}+(x+4)\cdot(x-4)-(x^2-16)-(3+|x|)$$

After:

$$x-1$$

demo derivative

Before:

$$\frac{x+1}{x+1}+\ln(e^x)+\sin(x)^2+\cos(x)^2+\sqrt{x^2}+\frac{2}{5}+\frac{7}{15}-\frac{13}{15}+(x+4)\cdot(x-4)-(x^2-16)-(3+|x|)$$

After:

$$1$$

demo integral of derivative

Before:

$$1$$

After:

$$x+C$$

Roots found (distinct):

Before:

$$x-1=0$$

After:

$$1$$

Restrictions summary:

$$\mathrm{unresolved}=0\,\mathrm{,}\,\mathrm{satisfied}=0\,\mathrm{,}\,\mathrm{conflicts}=0$$

7. Step Engine Showcase

Simplify trace (detail 1):

$$\begin{aligned} & \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + \sqrt{x^2} + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - (x^2-16) - (3+|x|) \\ &= \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + x + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - (x^2-16) - (3+x) \Rightarrow (x-4) \cdot \\ & (x+4) + 15 - x^2 + x \quad (\text{Complexity reduced} \mid \text{Identities: } \sin(u)^2 + \cos(u)^2 = 1; \ln(\exp(u)) = u) \\ &= (x-4) \cdot (x+4) + 15 - x^2 + x \Rightarrow x-1 \quad (\text{Expand then simplify}) \end{aligned}$$

Identities used (ordered):

$$1. \sin(u)^2 + \cos(u)^2 = 1$$

$$2. \ln(\exp(u)) = u$$

Active assumptions panel: active=0, satisfied=4, conflicts=0.

$$x+1 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$5 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$15 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$e^x > 0$$

$\ln(u)$ requires $u > 0$. [(satisfied, defined, fn:ln)]

Simplify trace (detail 4):

$$\begin{aligned} & \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + \sqrt{x^2} + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - (x^2-16) - (3+|x|) \\ &= \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + x + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - (x^2-16) - (3+x) \Rightarrow (x-4) \cdot \\ & (x+4) + 15 - x^2 + x \quad (\text{Complexity reduced} \mid \text{Identities: } \sin(u)^2 + \cos(u)^2 = 1; \ln(\exp(u)) = u) \\ &= (x-4) \cdot (x+4) + 15 - x^2 + x \Rightarrow x-1 \quad (\text{Expand then simplify}) \end{aligned}$$

Identities used (ordered):

$$1. \sin(u)^2 + \cos(u)^2 = 1$$

$$2. \ln(\exp(u)) = u$$

Active assumptions panel: active=0, satisfied=4, conflicts=0.

$$x+1 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$5 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$15 \neq 0$$

Division requires non-zero denominator. [(satisfied, defined, op:div)]

$$e^x > 0$$

$\ln(u)$ requires $u > 0$. [(satisfied, defined, fn:ln)]

Diff trace:

$$\begin{aligned} & \frac{d}{dx} \left((x^2+1)^3 \right) \\ &= \frac{d}{dx} \left((x^2+1)^3 \right) \Rightarrow 3(x^2+1)^{3-1} \cdot \frac{d}{dx}(x^2+1) \quad (\text{Apply power rule}) \\ &= \frac{d}{dx}(x^2+1) \Rightarrow \frac{d}{dx}(x^2) + \frac{d}{dx}(1) \quad (\text{Apply sum rule}) \\ & \quad \hookrightarrow \text{Branch 1} \\ & \quad = \frac{d}{dx}(x^2) \Rightarrow 2x^{2-1} \cdot \frac{d}{dx}(x) \quad (\text{Apply power rule}) \\ & \quad = 2x^{2-1} \cdot \frac{d}{dx}(x) \mapsto x \cdot 2 \quad (\text{Simplify derivative}) \\ & \quad \hookrightarrow \text{Branch 2} \\ & \quad = \frac{d}{dx}(1) \Rightarrow 0 \quad (\text{Differentiate}) \\ & \quad = \frac{d}{dx}(x^2) + \frac{d}{dx}(1) \mapsto x \cdot 2 \quad (\text{Simplify derivative}) \end{aligned}$$

$$= 3(x^2 + 1)^{3-1} \cdot \frac{d}{dx}(x^2 + 1) \mapsto x \cdot (x^2 + 1)^2 \cdot 6 \quad (\text{Simplify derivative})$$

Active assumptions panel: active=0, satisfied=0, conflicts=0.

U-sub integrate trace:

$$\int 2x \cdot \cos(x^2) \, dx$$

$$\text{set } u_1 = x^2$$

$$\text{derive } du_1 = x \cdot 2$$

$$= \int 2x \cdot \cos(x^2) \, dx \Rightarrow \int \cos(u_1) \, du_1 \quad (\text{Apply u-substitution})$$

$$= \int \cos(u_1) \, du_1 \mapsto \sin(u_1) + C \quad (\text{Integrate in } u_1)$$

$$= \sin(u_1) + C \Rightarrow \sin(x^2) + C \quad (\text{Back-substitute})$$

Active assumptions panel: active=0, satisfied=0, conflicts=0.

Additive integrate trace:

$$\int \sec(x)^2 + \csc(x)^2 \, dx$$

$$= \int (\csc(x)^2 + \sec(x)^2) \, dx \Rightarrow \int \csc(x)^2 \, dx + \int \sec(x)^2 \, dx \quad (\text{Apply linearity})$$

↳ Branch 1

$$= \int \csc(x)^2 \, dx \Rightarrow -\cot(x) + C \quad (\text{Apply square-family primitive})$$

↳ Branch 2

$$= \int \sec(x)^2 \, dx \Rightarrow \tan(x) + C \quad (\text{Apply square-family primitive})$$

$$= \int \csc(x)^2 \, dx + \int \sec(x)^2 \, dx \mapsto \tan(x) - \cot(x) + C \quad (\text{Simplify antiderivative})$$

Active assumptions panel: active=2, satisfied=0, conflicts=0.

$$\cos(x) \neq 0$$

sec(u) undefined when cos(u)=0. [(active, defined, fn:sec)]

$$\sin(x) \neq 0$$

csc(u) undefined when sin(u)=0. [(active, defined, fn:csc)]

Solve trace:

$$x^2 - 4 = 0$$

$$= x^2 - 4 - 0 \Rightarrow x^2 - 4 \quad (\text{Normalize to zero form})$$

Method: rational+numeric-complex

Detected polynomial degree: 2

Square-free part: $x^2 - 4$

where $a = 1$

where $b = 0$

where $c = -4$

Discriminant Δ : 16

$$= x^2 - 4 \mapsto (x - 2) \cdot (x + 2) \quad (\text{Factor form})$$

$$r_1 = 2$$

mult=1, exact=true, complex=false

$$r_2 = -2$$

mult=1, exact=true, complex=false

Depth override demo (detail:4 + depth:1):

$$\frac{d}{dx}((x^2 + 1)^3)$$

$$= \frac{d}{dx}((x^2 + 1)^3) \Rightarrow 3(x^2 + 1)^{3-1} \cdot \frac{d}{dx}(x^2 + 1) \quad (\text{Apply power rule})$$

$$= 3(x^2 + 1)^{3-1} \cdot \frac{d}{dx}(x^2 + 1) \mapsto x \cdot (x^2 + 1)^2 \cdot 6 \quad (\text{Simplify derivative})$$

Active assumptions panel: active=0, satisfied=0, conflicts=0.

8. Additional Examples (Expanded)

simplify (guarded cancel + reciprocal canonical):

Before:

$$\frac{x^2 - 1}{x - 1} + \frac{1}{\sin(x)}$$

After:

$$x + \csc(x) + 1$$

simplify (domain-sensitive):

Before:

$$\sqrt{x^2} + |x|$$

After:

$$x \cdot 2$$

diff (exp + inverse trig):

Before:

$$e^{3x} + \arctan(x^2)$$

After:

$$2 \left(x \cdot \left(\frac{1}{x^4 + 1} \right) \right) + 3e^x \cdot 3$$

diff (product + log):

Before:

$$x \cdot \ln(x)$$

After:

$$\ln(x) + 1$$

diff (non-smooth abs, piecewise-aware):

Before:

$$|x|$$

After:

$$\begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ \frac{d}{dx}(|x|) & \text{otherwise} \end{cases}$$

warnings:

$$1$$

diff (non-smooth min, piecewise-aware):

Before:

$$\min(x, x^2)$$

After:

$$\begin{cases} 1 & \text{if } x < x^2 \\ x \cdot 2 & \text{if } x^2 < x \\ \frac{d}{dx}(\min(x, x^2)) & \text{otherwise} \end{cases}$$

diff (non-smooth clamp, piecewise-aware):

Before:

$$\text{clamp}(x, 0, 1)$$

After:

$$\begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \wedge x < 1 \\ 0 & \text{if } x > 1 \\ \frac{d}{dx}(\text{clamp}(x, 0, 1)) & \text{otherwise} \end{cases}$$

integrate (by parts flavor):

Before:

$$x \cdot e^x$$

After:

$$x \cdot e^x - e^x + C$$

integrate (hyperbolic square-family):

Before:

$$\text{sech}(x)^2 + \text{csch}(x)^2$$

After:

$$\tanh(x) - \coth(x) + C$$

integrate (rational / partial fractions):

Before:

$$\frac{1}{x^2 - 1}$$

After:

$$\frac{\frac{-1}{2}}{x+1} + \frac{\frac{1}{2}}{x-1} + C$$

solve (cubic roots count):

Before:

$$x^3 - 6x^2 + 11x - 6$$

After:

$$3$$

solve (quartic complex roots count):

Before:

$$x^4 + 1$$

After:

$$4$$

factor:

Before:

$$x^4 - 16$$

After:

$$(x-2) \cdot (x+2) \cdot (x-2i) \cdot (x+2i)$$

taylor:
Before:

$$\sin(x)$$

After:

$$\frac{-1}{5040} \cdot x^7 + \frac{1}{120} \cdot x^5 + \frac{-1}{6} \cdot x^3 + x$$

limit:
Before:

$$\frac{\cos(x) - 1}{x}$$

After:

$$0$$

eval:
Before:

$$\sin\left(\frac{\pi}{2}\right) + \log_{10}(100)$$

After:

$$3$$

substitute:
Before:

$$x^2 + y^2, y \rightarrow 2x$$

After:

$$x^2 + (2x)^2$$

poly coeffs roundtrip:
Before:

$$x^4 - 3x + 2$$

After:

$$x^4 - 3x + 2$$

linear system:
Before:

$$x + y - 3, 2x - y$$

After:

$$x = 1, y = 2$$

domain display (t):

$$(-\text{inf}, -2] \cup (0, 4) \cup (4, \text{inf})$$

context simplify:
Before:

$$\frac{x^2 - 1}{x - 1}$$

After:

$$x + 1$$

context integrate:
Before:

$$\frac{1}{x^2-1}$$

After:

$$\frac{\frac{-1}{2}}{x+1}+\frac{\frac{1}{2}}{x-1}+C$$

8.1 Function + Identity Bundle
sign / sgn:
Before:

$$\text{sign}(-3) \text{ , } \text{sign}(0) \text{ , } \text{sign}(5)$$

After:

$$-1 \text{ , } 0 \text{ , } 1$$

floor + ceil + round:
Before:

$$\text{floor}(2.9) + \text{ceil}(2.1) + \text{round}(2.6)$$

After:

$$8$$

trunc / fracpart:
Before:

$$\text{trunc}(-2.9) \text{ , } \text{fracpart}(-2.9)$$

After:

$$-2 \text{ , } -0.8999999999999999$$

min / max / clamp:
Before:

$$\text{min}(3,4,1) \text{ , } \text{max}(3,4,1) \text{ , } \text{clamp}(10,0,5)$$

After:

$$1 \text{ , } 4 \text{ , } 5$$

clamp restriction metadata:
Before:

$$\text{clamp}(x,5,0)$$

After:

$$\text{conflicts}=1$$

identity expansion sample:
Before:

$$\frac{|x|}{x}+\frac{x}{|x|}+\min(x,x)$$

After:

$$x+2\text{sign}(x)$$

9. Translation Layer Note (compact)

v1 simplify alias:

$$1$$

v1 implicit diff alias:

$$-\left(\frac{y}{x}\right)$$

Canonical `cas.*` API is preferred for new documents.

10. Final Status

All unified checks passed.